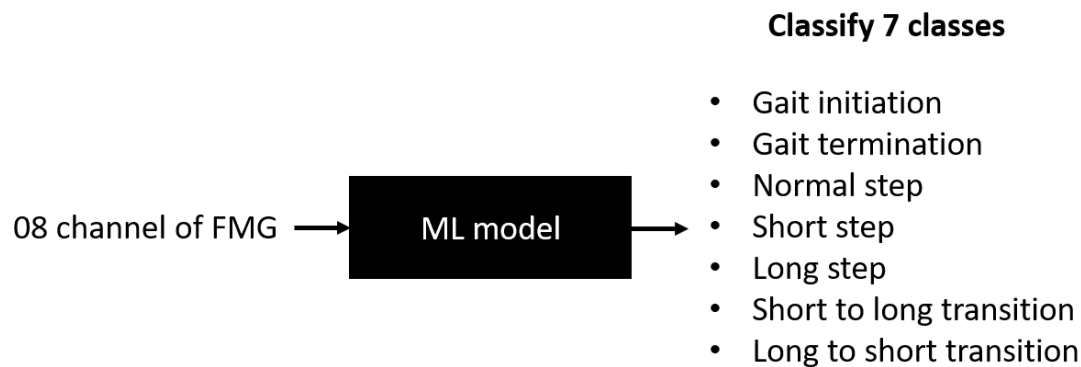


Planning two studies from the FMG data collected.

Study 1: Force myography-based classification of the gait initiation, termination, step length and transition for application of assistive device control

Objective: To develop ML model to classify the 7 classes



Data description:

360 steady state gait cycles for each of normal, long and short step length.

SI = 20 gait cycles

SCI = 20

LCT = 20

LT = 20

LI = 20

LCI = 20

SCI = 20

ST = 20

S_L = 20

S_L_C = 20

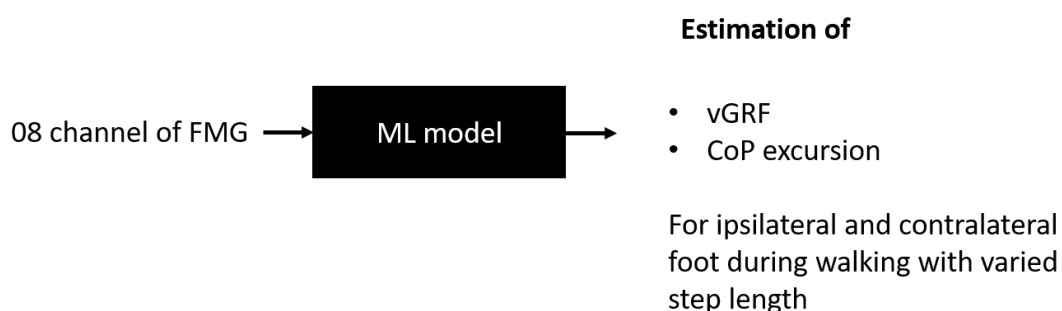
L_S = 20

L_S_C = 20

Study 2: Force myography-based estimation of ground reaction force and center of pressure excursion during gait.

Introduction:

Objective: To develop ML model to estimate vGRF and CoP excursion profile from the thigh force myography bands consisting of 8 channels data. The estimation will be done separately for three walking conditions i.e., normal step length, long step length and short step length.



Data description:

We have approximately **360** gait cycles of the FMG, vGRF, CoP excursion collected for each of normal, long and short step length for **08** healthy and **02** amputee population. For amputees 160 gait cycles for the amputated and 160 for the intact limb will be there.

This data was collected over 40 trials during overground walking.

The data is properly labelled as follows

For each subject you will find a total of 06 files. 'X' denotes the subject

- X_1** -> Trials 1 to 20 for both the limb. This file consists of the time (C2), Trigger (begin and end of trial, C3), then C4 to C11 is left band FMG channels and C12 to C19 is the FMG right band channel.
- X_1L** -> Insole data (vGRF and CoP excursion) for the left foot for the trial 1 to 20. This file consists of the time (C2), Trigger (begin and end of trial, C3), then C4 and C5 corresponds to the CoP excursion and vGRF.
- X_1R** -> Insole data (vGRF and CoP excursion) for the right foot for the trial 1 to 20. This file consists of the time (C2), Trigger (begin and end of trial, C3), then C4 and C5 corresponds to the CoP excursion and vGRF.
- X_21** -> Trials 21 to 40 for both the limb. This file consists of the time (C2), Trigger (begin and end of trial, C3), then C4 to C11 is left band FMG channels and C12 to C19 is the FMG right band channel.
- X_21L** -> Insole data (vGRF and CoP excursion) for the left foot for the trial 21 to 40. This file consists of the time (C2), Trigger (begin and end of trial, C3), then C4 and C5 corresponds to the CoP excursion and vGRF.
- X_21R** -> Insole data (vGRF and CoP excursion) for the right foot for the trial 21 to 40. This file consists of the time (C2), Trigger (begin and end of trial, C3), then C4 and C5 corresponds to the CoP excursion and vGRF.

The Data collection protocol.

